

Responding to the chikungunya virus in west Africa

Chikungunya, a neglected viral disease transmitted by *Aedes* mosquitoes, which also carry dengue and yellow fever, has been identified in more than 110 countries across Asia, Africa, Europe, and the Americas.¹ Although chikungunya typically results in low mortality, the disease can lead to prolonged illness and, in rare instances, complications, such as neurological disorders, eye inflammation, and heart issues.² No specific treatment or vaccine is available, making prevention and symptomatic treatment the primary management strategies.

From 2017 to 2022 we have seen a resurgence of chikungunya in various regions.² In 2023, more than 460 000 cases and 350 deaths were reported globally, with the highest burden of the disease in South and central America.² This resurgence is also notable in west Africa, with 565 confirmed cases across Senegal, Burkina Faso, Mali, The Gambia, and Guinea by November, 2023. The re-emergence in regions such as southeast Senegal (Kédougou), paralleling the biggest rise in local temperatures between the 1980s and 2023 and concurrent dengue outbreaks, suggests environmental, socioeconomical, and anthropical factors contributing to virus spillover from sylvatic to urban cycles. The spread of *Aedes albopictus*, possibly aided by climate change, increasing trade, uncontrolled urbanisation, and increased global connectivity,³ has been a contributing factor, as shown in past outbreaks in Cameroon, Gabon, and the Democratic Republic of the Congo.⁴

Addressing chikungunya in west African countries, which vary in resources and infrastructure, presents several challenges. These challenges include poor access to diagnostics and inadequate funding for research and development. Resolving many of these issues might require additional support

and collaboration from the international community. To address these issues, promoting domestic funding for sustainability and enhancing innovation and health technology access through the development of digital platforms and integrated case management strategies is crucial. This effort can include mapping data from endemic countries. Training local clinicians, investing in public health laboratories and local manufacturing hubs, implementing genomic surveillance, and developing and delivering innovative and affordable point-of-care solutions are also key steps. These measures, aligned with the One Health approach, can improve community-based surveillance and risk assessment, fostering cross-country collaboration and health sector cooperation.

Another key aspect is the accelerated confirmation of clinical trials for potential vaccines, such as the IxChiq vaccine (Valneva, France), which received US Food and Drug Administration approval after phase 3 trials.⁵ Supporting vaccine manufacturing in low-income and middle-income countries is essential for a rapid and equitable response to future outbreaks. This collective approach, embracing technology, health-care infrastructure improvements, and international cooperation, is key for managing chikungunya effectively and bolstering global health resilience.

We declare no competing interests. Opinions, interpretations, conclusions, and recommendations are those of the authors and are not necessarily endorsed by the US Army.

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Cofinancing immunisation through national health insurance

Sustainable immunisation financing is a global health issue of utmost concern that requires serious political attention and commitment from all countries. However, the efforts that many low-income and middle-income countries (LMICs), with persistently underfunded immunisation systems, are making to institute policy reforms that can stimulate more domestic immunisation financing are alarmingly passive. The COVID-19 pandemic exposed the gross level of unpreparedness of many countries to meet the needs of their people during external shocks (eg, unprecedented health emergencies), thus underscoring the importance of adequately funded and self-reliant health systems. Yet, many countries continue to depend on donor funds for routine immunisation, even in the post-pandemic period, without any tangible and viable action plan to transition away from this support. Many of the LMICs that are funded by Gavi, the Vaccine Alliance, are either in the preparatory or accelerated transition phase and, if that period ends, they will need to become fully self-financed.¹

The possibility of securing higher allocations for immunisation in national budgets is currently capricious because countries are confronting an unprecedented level of

For more on the Immunization Agenda 2030 Framework for Action see <https://www.immunizationagenda2030.org/framework-for-action>

complex epidemiological transitions, macroeconomic issues, and political instability, all of which constrain their resources. Without sufficient and predictable financing, many LMICs will not make any substantial progress towards achieving the goals of the Immunization Agenda 2030 Framework for Action. Failing to meet these goals would have negative effects on the lives and wellbeing of all people.

A *Lancet* Correspondence by Uchenna A Amaechi and colleagues² suggested that countries can further enhance their fiscal space for immunisation by using innovative cofinancing mechanisms. A feasible opportunity that should be urgently explored is national health insurance, a health financing option that many countries are adopting to drive progress towards universal health coverage.³ Although coverage varies, some countries already have substantial pools, with their national health insurance generating revenues through contributions from individuals themselves or through alternative channels for those who are unable to pay.⁴ With the concerted effort of crucial stakeholders, national health insurance can be galvanised to cofinance primary health care, most pertinently, immunisation services, including vaccines. Redirecting pooled funds to immunisation would require the development and implementation of multiple policy reforms with sector-wide implications. This call to action is intended to stimulate researchers, donors, policy makers, and programme implementers to rethink and begin to redesign the financing model for primary health care and immunisation, especially in LMICs, in a manner that positions national health insurance as a potential important funding source.

Our key recommendations for action to cofinance immunisation through national health insurance are: to specify routine immunisation, including vaccines, in the health benefit package of national health insurance schemes; to establish contract agreements with national immunisation programmes

to procure and supply recommended vaccines through existing mechanisms on behalf of children and adults that are covered under national health insurance schemes; to implement a prospective input-and-output-based payment method for immunisation services, including vaccines; and to deploy an enhanced immunisation performance monitoring system to track and link provider incentive payments with health-care facility functionality, immunisation service utilisation, and immunisation service quality.

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One Health pandemic preparedness: the role of companion animals in disease transmission

Urbanisation creates novel interfaces that facilitate cross-species

transmission of emerging infectious diseases. Outwardly, so-called city encroachment into natural areas creates periurban interfaces between people, livestock, and wildlife. Inwards, companion animals are fully integrated into urban life, creating so-called animal encroachment into dwellings and novel human-animal interfaces, with the potential for complex urban multi-species transmission networks.

Historically, the urban interface linking humans, companion animals, and urban wildlife has received limited attention in pandemic preparedness and response. However, recent events highlight its importance: several companion dogs and cats were infected with SARS-CoV-2 by their owners, and at least one human COVID-19 outbreak originated from pet hamsters.¹ Most recently, companion cats in Poland and South Korea and dogs in France and Canada were infected with avian influenza H5N1 clade 2.3.4.4b following outbreaks in urban wild and domestic birds.²

WHO, the World Organisation for Animal Health, the Food and Agriculture Organization, and the UN Environment Programme have jointly called for the prioritisation and acceleration of One Health, an integrated, unifying approach that aims to sustainably balance and optimise the health of people, animals, and ecosystems.³ Through this collaboration, they aim to build intersectoral workforces that “strengthen and sustain prevention of pandemics and health threats at source, targeting activities and places that increase the risk of zoonotic spillover between animals to humans”.⁴ Aligned with this call, WHO published the Mosaic Respiratory Surveillance Framework,⁵ highlighting the importance of community event-based surveillance (CEBS) in targeting at-risk populations at the human-animal interface. Although CEBS is deemed essential for rapidly detecting,